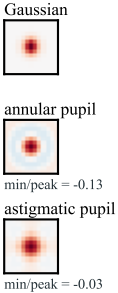
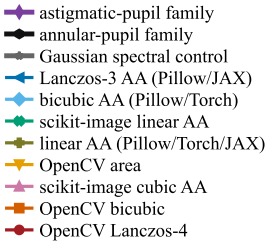
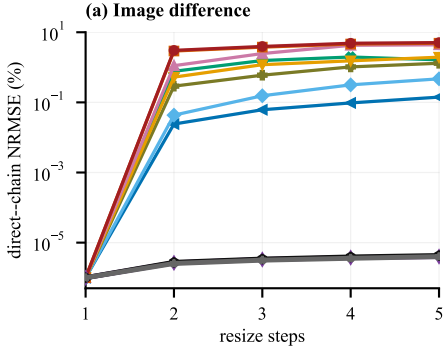
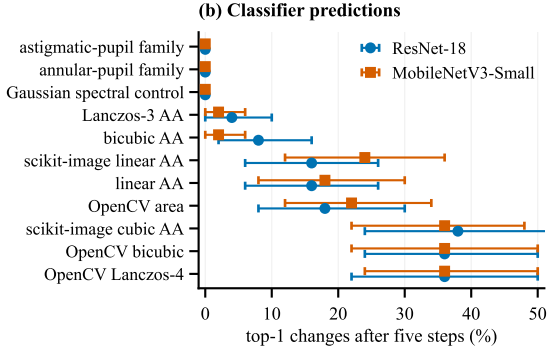

Gaussian
annular pupil
min/peak = -0.13
astigmatic pupil
min/peak = -0.03


astigmatic-pupil family
annular-pupil family
Gaussian spectral control
Lanczos-3 AA (Pillow/JAX)
bicubic AA (Pillow/Torch)
scikit-image linear AA
linear AA (Pillow/Torch/JAX)
OpenCV area
scikit-image cubic AA
OpenCV bicubic
OpenCV Lanczos-4


(a) Image difference
direct-chain NRMSE (%)
10^{1}
10^{-1}
10^{-3}
10^{-5}
1
2
3
4
5
resize steps


(b) Classifier predictions
ResNet-18
MobileNetV3-Small
astigmatic-pupil family
annular-pupil family
Gaussian spectral control
Lanczos-3 AA
bicubic AA
scikit-image linear AA
linear AA
OpenCV area
scikit-image cubic AA
OpenCV bicubic
OpenCV Lanczos-4
0
10
20
30
40
50
top-1 changes after five steps (%)